

Computer Simulation II - Summary Report

Schewtschik, Guilherme (3748052)

July 6, 2026

Abstract

This report presents a collection of computational investigations in statistical physics and stochastic simulation methods.

1 Introduction

This report is based on the Computer Simulation II lecture course at the University of Leipzig. Covering mainly the simulation of stochastic process related to the Ising-Potts and XY models, with main focus on cluster algorithms, detection of phase transitions, finite size scaling of observables, and the analysis of critical exponents. The report is structured in a way that each section corresponds to a specific topic covered in the course, providing a summary of the methods used, results obtained, and conclusions drawn from the simulations.

2 Cluster Algorithms for the Ising Model

Using the foundations established by the Metropolis algorithm [7] one may naturally be interested in the behaviour of the model around the critical temperature, at a first attempt using the metropolis algorithm one may find that for systems of relevant size the simulation becomes quickly impractical, with run times exploding and errors accumulating. This is due to the fact that local update algorithms such as the Metropolis algorithm suffer from critical slowing down [6] near the critical temperature, where the systems becomes highly correlated, i.e.: local changes affect the whole system in significant ways. To overcome this major limitation we look to the cluster algorithms, that aim to overcome this problem by using the high autocorrelation of the model to their advantage.

The first prolific Cluster algorithm was the Swendsen-Wang algorithm [9], which significantly improved the performance of the simulations but still suffered from critical slowing down in higher dimensions. The pipeline of the Swendsen-Wang algorithm is as follows:

- given a spin configuration establish bonds between neighboring parallel spins with probability $p = 1 - e^{-2\beta J}$
- for each individual cluster, flip the spins with probability 1/2.
- repeat the process for a number of iterations, measuring observables after each iteration.

We can see where the shortcomings of the Swendsen-Wang algorithm are, as it flips all clusters at once, which can lead to a high rejection rate and slow convergence.

2.1 Wolff Algorithm

The Wolff algorithm is our main focus when it comes to clusters algorithms, as it improves upon the Swendsen-Wang algorithm by only flipping a single cluster of spins at a time, rather than all clusters. This leads to a more efficient exploration of the configuration space, especially near the critical temperature.

The pipeline of the Wolff algorithm [10] is as follows:

- given a spin configuration, randomly select a seed spin and establish bonds between neighboring parallel spins with probability $p = 1 - e^{-2\beta J}$, adding them to the cluster.
- repeat the process for each newly added spin until no more spins can be added to the cluster.
- flip the entire cluster of spins.
- repeat the process for a number of iterations, measuring observables after each iteration.

So now all the highly correlated spin clusters are flipped at once, largely avoiding the problem of critical slowing down, and allowing for a more efficient exploration of the configuration space.

To check this empirically we can check the integrated autocorrelation time of the magnetization, that is:

$$\tau_{int} = \frac{1}{2} + \sum_{t=1}^{\infty} \frac{\langle m(t)m(0) \rangle - \langle m(0) \rangle^2}{\langle m(0)^2 \rangle - \langle m(0) \rangle^2} \quad (2.1)$$

computing the integrated magnetization autocorrelation time for different system sizes around the critical temperature, for the Wolff algorithm we get:

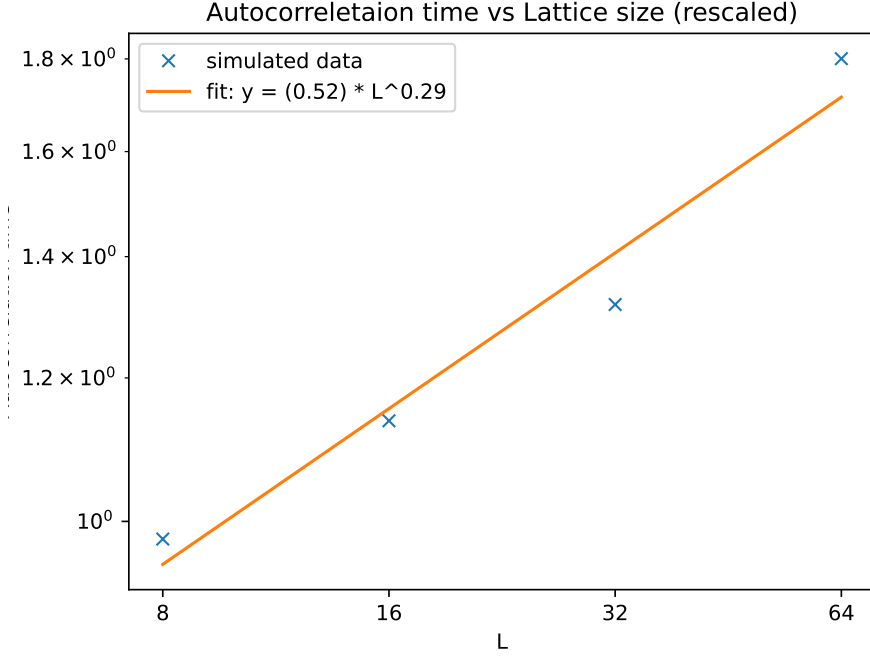


Figure 1: Magnetization autocorrelation time in units of sweeps

so we can see that the Wolff algorithm has the autocorrelation time scaling as $\tau_{int} \sim L^{0.29}$, which is a significant improvement over the Metropolis algorithm, which scales as $\tau_{int} \sim L^{2.1}$ [6].

2.2 Improved estimators

Cluster algorithms also allow us to use improved estimators, that is, we can use the geometry of the system and the clusters to better estimate the observables we are interested in, be that a faster convergence or a lower error estimate. The classical example for this is the magnetization, first defined as:

$$m = \frac{1}{N} \sum_{i=1}^N s_i \quad (2.2)$$

That is basically the average of the spins, but since the clusters already tell us how the spins are distributed we can calculate the magnetization by just looking at the average size of the clusters [10], that is:

$$m = \frac{1}{N} \sum_c n_c s_c \quad (2.3)$$

where n_c is the size of cluster c and s_c is the spin of the cluster. As the number of clusters is much smaller than the number of spins, this leads to a significant reduction in the error of the magnetization estimate. this is a general feature of cluster algorithms, that they allow for improved estimators for many observables, leading to a more efficient exploration of the configuration space.

We can easily see the impact of improved estimators with the correlation function where the correlation function is defined as:

$$G(r) = \langle s_i s_j \rangle - \langle s_i \rangle \langle s_j \rangle \quad (2.4)$$

With cluster estimator [10] given by:

$$G(r) = \frac{1}{N} \sum_c n_c^2 - \left(\frac{1}{N} \sum_c n_c s_c \right)^2 \quad (2.5)$$

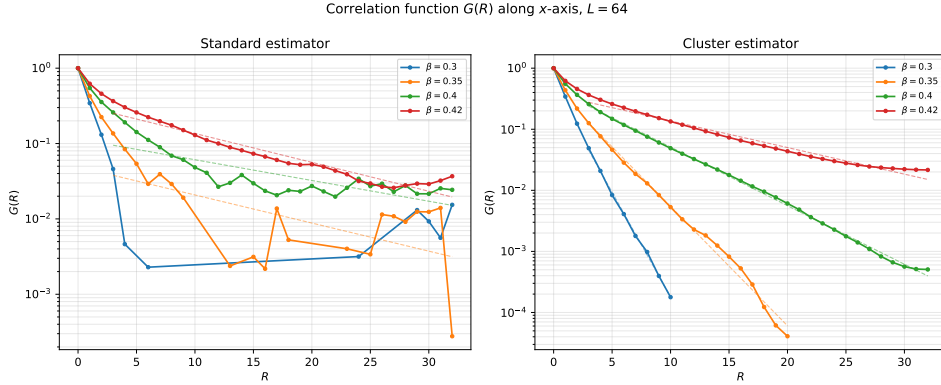


Figure 2: Correlation function for the normal and improved estimators

We can see that the improved estimator shows significantly less noise than the normal estimator, making the data actually usable.

3 Non-Ising models

3.1 The Potts model

The Potts model [8] is one of the simplest generalizations of the Ising model, where instead of only 2 spin possibilities, $s = \pm 1$ we introduce q possible spins.

We may illustrate this at the most simple case of $q = 3$, at the critical temperature $\beta_c = \ln(1 + \sqrt{q})$ [1]:

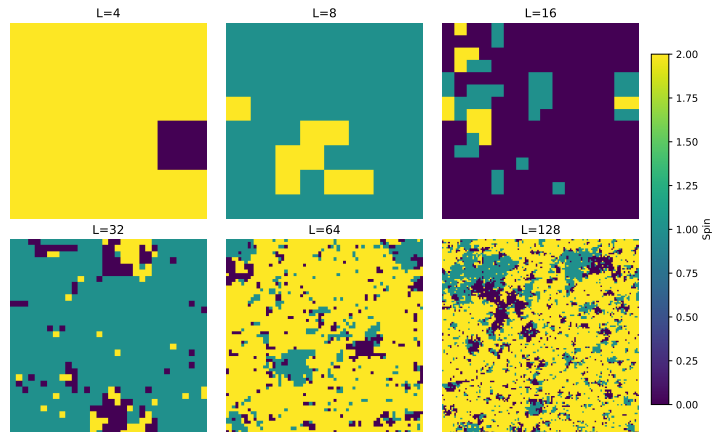


Figure 3: Potts model with $q = 3$ at the critical temperature

The Potts model can also be simulated using cluster algorithms, for this we may use a modified version of the Wolff algorithm, where the cluster construction procedure is the same, just having an altered probability of adding a spin to the cluster, $p_{add} = 1 - e^{-\beta J}$. and since flipping the spins is no longer well defined one just randomly selects a new spin in q and assigns that to the cluster.

3.2 The XY model

The XY model [2, 5] is further more a continuous generalization of the potts model, where now the spins don't assume discrete values but instead they can take any value in a unit 2-sphere, that is, the spins are now vectors of form $\vec{s} = (\cos(\theta), \sin(\theta))$ with $\theta \in [0, 2\pi)$.

With this we need to present a new hamiltonian in order to be able to simulate the model, that is [2, 5]:

$$H = -J \sum_{\langle i,j \rangle} \vec{s}_i \cdot \vec{s}_j = -J \sum_{\langle i,j \rangle} \cos(\theta_i - \theta_j) \quad (3.1)$$

With that in hands we may want to use a cluster algorithm (e.g.: Wolff algorithm) to simulate the XY model, but due to how the spins are continuous we would practically have no clusters if taken the naive approach of just adding parallel spins to the cluster, instead how we proceed is to randomly select a unit vector to which we project the spins [10], taking as back to a one dimensional problem, just now continuous, and then we may proceed normally with the Wolff algorithm, adding spins to the cluster if they are parallel to the selected vector.

For illustration, we can see how our standard observables behave in the XY model:

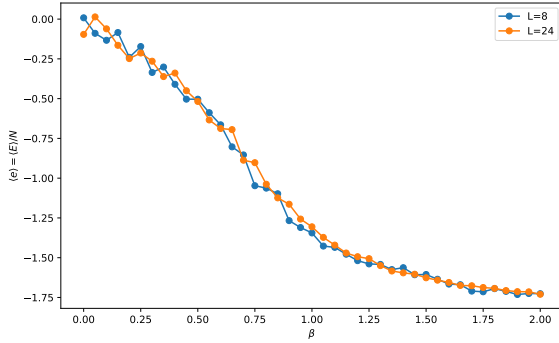


Figure 4: Mean energy per site $\times \beta$.

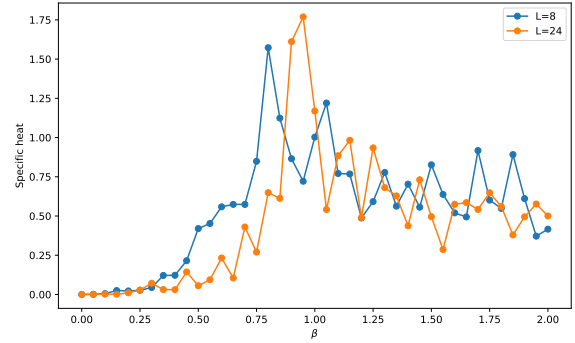


Figure 5: Specific heat $\times \beta$.

4 Finite Size Scaling and critical exponents

The Finite Size Scaling (FSS) approach is a tool used to study the behavior of the system near the critical point, it's main idea is to use the fact that the correlation length diverges at the critical point, and that for finite systems the correlation length is limited by the system size, leading to a rounding of the singularities in the observables. For this it is also useful to define the critical exponents, they define how the observables behave near the critical point, a short list of common critical exponents is [6]:

- specific heat exponent, $C \sim |T - T_c|^{-\alpha}$
- magnetization exponent, $m \sim |T - T_c|^\beta$
- susceptibility exponent, $\chi \sim |T - T_c|^{-\gamma}$
- correlation length exponent, $\xi \sim |T - T_c|^{-\nu}$

Now we can introduce the FSS ansatz [3]:

$$O(T, L) = L^{\kappa/\nu} \tilde{O} \left(\left| \frac{T - T_c}{T_c} \right| L^{1/\nu} \right) \quad (4.1)$$

Where $O(T, L)$ is some observable of interest. We can take this further and get the scaling of the maximum of the observable [3], that is:

$$O_{\max}(L) \sim L^{\kappa/\nu} \quad (4.2)$$

This allows us to extract the critical exponents from the simulations, by fitting the data to the FSS ansatz, and also later allows us to estimate the critical temperature.

4.1 Determining the critical exponents

Through the course we calculated exact critical exponents getting the following results:

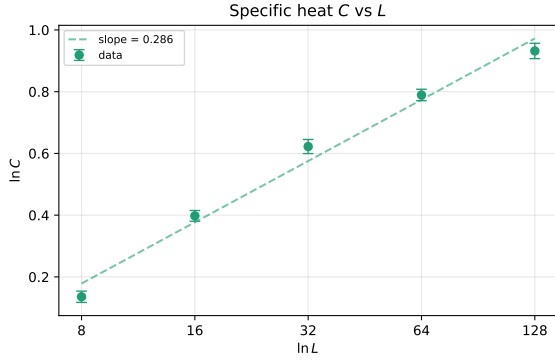


Figure 6: Specific heat.

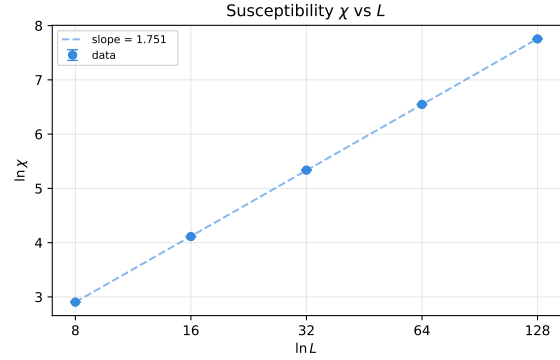


Figure 7: Susceptibility.

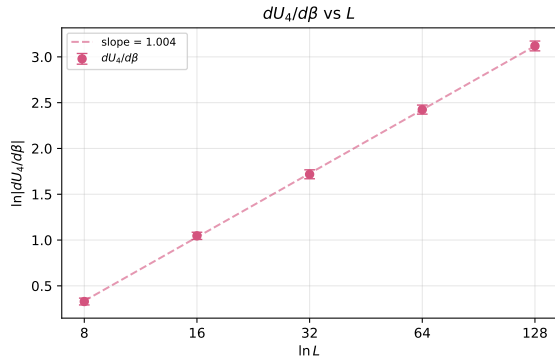


Figure 8: $\frac{dU_4}{d\beta}$

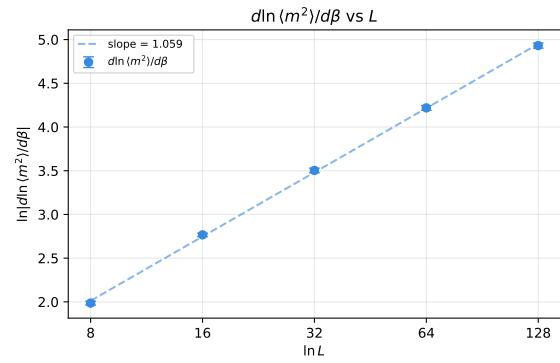


Figure 9: $\frac{d \ln m_2}{d\beta}$

So we get:

$$\begin{aligned}
 C_{\max} &\sim L^{\alpha/\nu} \sim L^{0.286} \\
 \chi_{\max} &\sim L^{\gamma/\nu} \sim L^{1.751} \\
 \frac{dU_4}{d\beta} &\sim L^{1/\nu} \sim L^{1.004} \\
 \frac{d \ln m_2}{d\beta} &\sim L^{1/\nu} \sim L^{1.059}
 \end{aligned}$$

5 Phase transitions

To detect phase transitions in the Ising model there are several methods, the ones explored in the course are the Binder cumulant, specific heat and susceptibility.

5.1 Binder cumulant

The Binder cumulant [3] is a dimensionless quantity that is used to detect phase transitions, it is defined as:

$$U_4 = 1 - \frac{\langle m^4 \rangle}{3\langle m^2 \rangle^2} \quad (5.1)$$

In order to understand the behaviour of the cumulant around the critical temperature we must first analyse the behaviour at the thermodynamical limits, $T \rightarrow 0$ and $T \rightarrow \infty$, at $T \rightarrow 0$ the system is in a ferromagnetic state, with all spins aligned having the maximum magnetization possible, leading to $\langle m^4 \rangle = \langle m^2 \rangle^2$, and thus $U_4 = 2/3$. In the other hand as $T \rightarrow \infty$ the system is in a paramagnetic state, with all spins randomly oriented, leading to $\langle m^4 \rangle = 3\langle m^2 \rangle^2$ [3], and thus $U_4 = 0$. So the cumulant first starts at $2/3$ and then goes to 0, what one may expect is that the cumulant will present a sigmoidal like behaviour, becoming zero at the critical temperature.

We may further analyze the behaviour of the cumulant by looking at its FSS ansatz, from equation 4.1 we have:

$$U_4(T, L) = \tilde{U}_4 \left(\left| \frac{T - T_c}{T_c} \right| L^{1/\nu} \right) \quad (5.2)$$

But also we may write down the FSS ansatz for the powers of the magnetization:

$$m^4(T, L) = L^{-4\beta/\nu} \tilde{m}^4 \left(\left| \frac{T - T_c}{T_c} \right| L^{1/\nu} \right) \quad (5.3)$$

$$m^2(T, L) = L^{-2\beta/\nu} \tilde{m}^2 \left(\left| \frac{T - T_c}{T_c} \right| L^{1/\nu} \right) \quad (5.4)$$

Combining equations 5.2, 5.3 and 5.4 we get:

$$U_4(T, L) = 1 - \frac{L^{-4\beta/\nu} \tilde{m}^4 \left(\left| \frac{T - T_c}{T_c} \right| L^{1/\nu} \right)}{3L^{-4\beta/\nu} \tilde{m}^2 \left(\left| \frac{T - T_c}{T_c} \right| L^{1/\nu} \right)^2} = 1 - \frac{\tilde{m}^4 \left(\left| \frac{T - T_c}{T_c} \right| L^{1/\nu} \right)}{3\tilde{m}^2 \left(\left| \frac{T - T_c}{T_c} \right| L^{1/\nu} \right)^2} \quad (5.5)$$

So at the critical temperature we have that $U_4(T_c, L) = 1 - \frac{\tilde{m}^4(0)}{3\tilde{m}^2(0)^2}$ [3], which is a constant independent of the system size, and thus we can use the intersection of the cumulant curves for different system sizes to estimate the critical temperature.

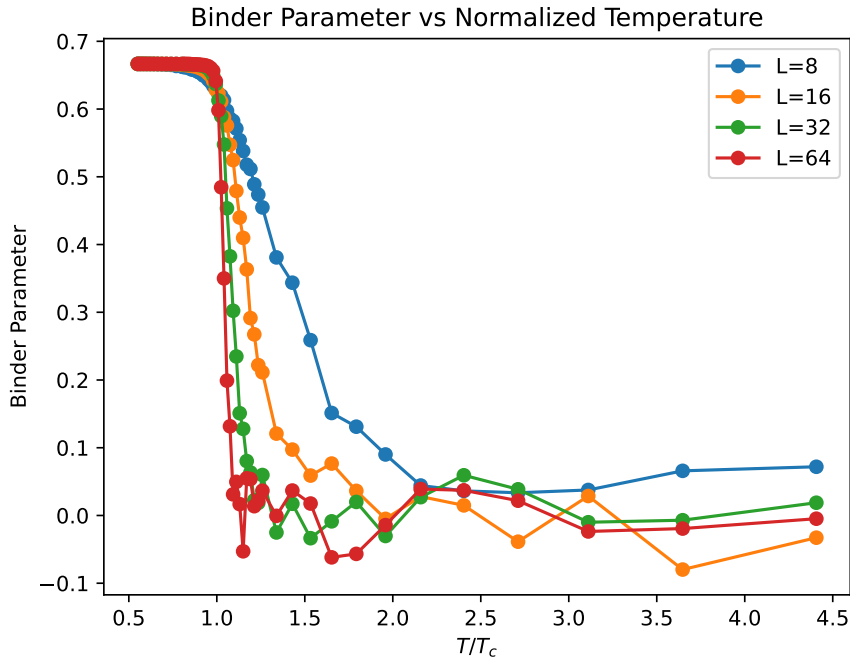


Figure 10: Binder cumulant for different system sizes.

5.2 Susceptibility and Specific heat as Phase transition detectors

For the Susceptibility and specific heat we can determine the critical temperature by determining where the maximum occurs for different system sizes [3].

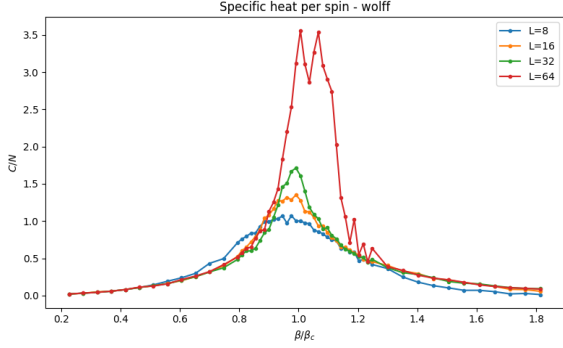


Figure 11: Specific heat for different system sizes.

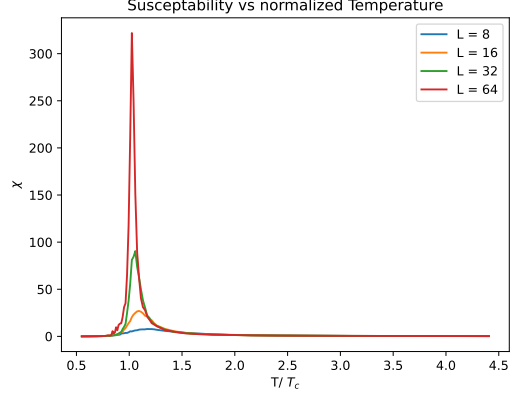


Figure 12: Susceptibility for different system sizes.

6 Single and Multiple Histogram reweighting

The histogram reweighting method is a technique that consists on extrapolating the results of a simulation on a few, or at a singular, temperature to a range of temperatures.

6.1 Single Histogram Reweighting

For a given energy histogram $H_0(E)$ obtained from a simulation at temperature T_0 , we can reconstruct a given observable near the temperature T_0 by reweighting the histogram [4]:

$$\langle O \rangle_T = \frac{\sum_E O(E) H_0(E) e^{-(\beta - \beta_0)E}}{\sum_E H_0(E) e^{-(\beta - \beta_0)E}} \quad (6.1)$$

This technique is although limited to a small range of temperatures where the correlation between histograms is high, we know the autocorrelation length for the energy so using the FSS ansatz one can determine that in order for the reweighting to be valid the condition $\Delta\beta \sim L^{-1/\nu} \sim 1/\sigma_E$ [4], where σ_E is the standard deviation of the energy histogram given by $\sigma_E = \sqrt{\langle E^2 \rangle - \langle E \rangle^2} = \sqrt{k_B T^2 C}$.

6.2 Multiple Histogram reweighting

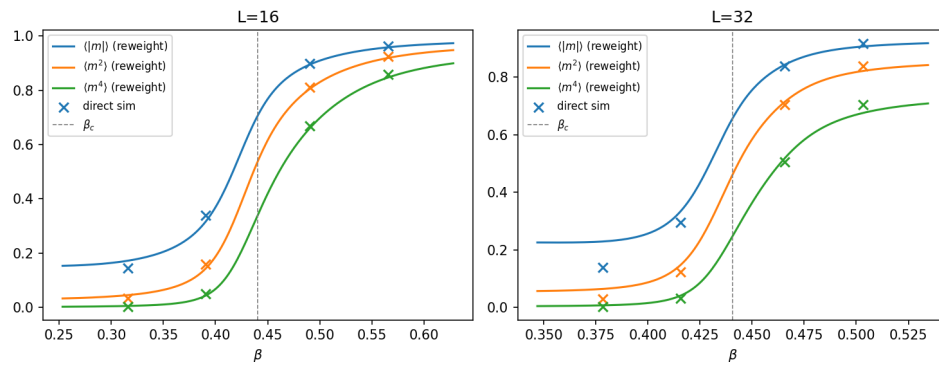
For the multiple histogram reweighting we now use the histograms to estimate the global distribution of states [4]:

$$\Omega(E) = \frac{\sum H_i(E)}{\sum N_i e^{-\beta E - f_i}} \quad (6.2)$$

Where N_i is the number of measurements of the histogram and $f_i = \ln(Z(\beta_i))$ so we get the observable $\langle O \rangle$:

$$\langle O \rangle_T = \frac{\sum_E O(E) \Omega(E) e^{-\beta E}}{\sum_E \Omega(E) e^{-\beta E}} \quad (6.3)$$

Using this for the different powers of magnetization:



So we can see that the reweighted results agree well with the simulated results.

References

- [1] Rodney J. Baxter. *Exactly Solved Models in Statistical Mechanics*. Academic Press, 1982.
- [2] V. L. Berezinskii. “Destruction of Long-range Order in One-dimensional and Two-dimensional Systems Having a Continuous Symmetry Group”. In: *Soviet Journal of Experimental and Theoretical Physics* 32 (1971), pp. 493–500.
- [3] Kurt Binder. “Finite Size Scaling Analysis of Ising Model Block Distribution Functions”. In: *Zeitschrift für Physik B* 43 (1981), pp. 119–140. DOI: 10.1007/BF01293604.
- [4] Alan M. Ferrenberg and Robert H. Swendsen. “New Monte Carlo Technique for Studying Phase Transitions”. In: *Physical Review Letters* 61 (1988), pp. 2635–2638. DOI: 10.1103/PhysRevLett.61.2635.
- [5] J. M. Kosterlitz and David J. Thouless. “Ordering, Metastability and Phase Transitions in Two-Dimensional Systems”. In: *Journal of Physics C* 6 (1973), pp. 1181–1203. DOI: 10.1088/0022-3719/6/7/010.
- [6] David P. Landau and Kurt Binder. *A Guide to Monte Carlo Simulations in Statistical Physics*. 5th ed. Cambridge University Press, 2021. DOI: 10.1017/9781108776843.
- [7] Nicholas Metropolis et al. “Equation of State Calculations by Fast Computing Machines”. In: *Journal of Chemical Physics* 21.6 (1953), pp. 1087–1092. DOI: 10.1063/1.1699114.
- [8] Renfrey B. Potts. “Some Generalized Order-Disorder Transformations”. In: *Proceedings of the Cambridge Philosophical Society* 48 (1952), pp. 106–109. DOI: 10.1017/S0305004100027419.
- [9] Robert H. Swendsen and Jian-Sheng Wang. “Nonuniversal Critical Dynamics in Monte Carlo Simulations”. In: *Physical Review Letters* 58 (1987), pp. 86–88. DOI: 10.1103/PhysRevLett.58.86.
- [10] Ulli Wolff. “Collective Monte Carlo Updating for Spin Systems”. In: *Physical Review Letters* 62 (1989), pp. 361–364. DOI: 10.1103/PhysRevLett.62.361.